

1. Prim's Algorithm (Minimum Spanning Tree)

Theory:

Prim's algorithm is a greedy algorithm that finds a minimum spanning tree for a weighted undirected graph. It starts with an arbitrary node and grows the spanning tree by adding the smallest possible edge that connects the tree to a new vertex.

Steps:

1. Initialize a set of visited nodes and an empty minimum spanning tree.
 2. Select an arbitrary starting node.
 3. While there are unvisited nodes, add the smallest edge that connects a visited node to an unvisited node.
 4. Repeat until all nodes are included.
-

2. Kruskal's Algorithm (Minimum Spanning Tree)

Theory:

Kruskal's algorithm is another greedy algorithm used to find the minimum spanning tree of a graph. It sorts all the edges of the graph in increasing order of their weights and adds edges to the spanning tree in that order, ensuring no cycles are formed.

Steps:

1. Sort all edges by weight.
 2. Initialize the minimum spanning tree as empty.
 3. For each edge, if adding it doesn't create a cycle, add it to the tree.
 4. Repeat until the tree spans all vertices.
-

3. Breadth-First Search (BFS)

Theory:

BFS is an algorithm for traversing or searching tree or graph data structures. It starts at the root (or any arbitrary node) and explores all the neighbor nodes at the present depth before moving on to nodes at the next depth level.

Steps:

1. Start at the source node.
 2. Visit all unvisited neighbors and add them to a queue.
 3. Dequeue and visit the next node from the front of the queue.
 4. Repeat until all nodes are visited.
-

4. Depth-First Search (DFS)

Theory:

DFS is another graph traversal algorithm, but unlike BFS, it explores as far as possible along each branch before backtracking.

Steps:

1. Start at the source node.
 2. Visit a node and recursively visit all its unvisited neighbors.
 3. Backtrack when no unvisited neighbors are found.
 4. Repeat for all nodes.
-

5. A Algorithm (Pathfinding)*

Theory:

A* is a popular pathfinding algorithm used in many applications, such as GPS navigation. It combines the features of BFS and Dijkstra's Algorithm, using heuristics to guide the search toward the goal node.

Steps:

1. Initialize a priority queue (open list) that stores nodes along with their costs.
 2. Evaluate the cost of moving from the start node to each neighboring node.
 3. Use a heuristic to estimate the cost to reach the goal from the neighboring nodes.
 4. Continue expanding the node with the lowest total cost (current cost + heuristic).
-

6. N-Queens Problem

Theory:

The N-Queens problem involves placing N queens on an N×N chessboard in such a way that no two queens threaten each other (no two queens can share the same row, column, or diagonal).

Approach:

- Backtracking is often used to solve this problem. You start by placing a queen in the first row and try placing subsequent queens row by row while checking for conflicts.
 - If a conflict arises, you backtrack and try placing the queen in a different position.
-

7. Graph Coloring

Theory:

Graph coloring is the assignment of labels (called colors) to the vertices of a graph such that no two adjacent vertices share the same color. This is typically used in scheduling and resource allocation problems.

Steps:

1. Start with the first vertex and assign a color.
 2. Move to the next vertex and assign the lowest color not used by its neighbors.
 3. Repeat until all vertices are colored.
-

8. Chatbot Technology

Overview:

A chatbot is a software application designed to simulate human conversation. They are widely used for customer service and other interactive tasks.

Types of Chatbots:

1. **Rule-based chatbots:** These operate based on predefined rules and scripts.
2. **AI-powered chatbots:** These use natural language processing (NLP) and machine learning to understand and generate responses.

Algorithms Involved:

- **NLP techniques** such as tokenization, named entity recognition (NER), and part-of-speech tagging.

- **Dialogue management** using intent classification and response generation.
-

9. EC2 (Elastic Compute Cloud)

Overview:

EC2 is a web service provided by Amazon Web Services (AWS) that offers scalable computing capacity in the cloud. EC2 allows users to rent virtual machines on demand, known as instances.

Features:

- Flexible and scalable infrastructure.
 - Pay-as-you-go pricing.
 - Wide variety of instance types based on computing, memory, and storage needs.
-

10. Google App Engine (GAE)

Overview:

GAE is a fully managed serverless platform for developing and deploying web applications. It allows developers to focus on writing code while Google manages the infrastructure.

Features:

- Automatic scaling and load balancing.
- Support for multiple programming languages (Python, Java, Go, etc.).
- Integration with Google Cloud services.